

MKT-06-CUSTOMER-SEGMENTS

MKT-06: Customer Segments and Persona Analysis

Project AEGIS / Carrington Storm Motors

Detailed Customer Persona Profiles — Buying Process, Pain Points, Objections

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EXECUTIVE SUMMARY

CSM addresses six primary customer segments, each with distinct procurement processes, decision criteria, pain points, and buying triggers. This document profiles each segment in detail to guide product development, sales strategy, and marketing messaging. Segments are ordered by anticipated revenue contribution over the first 5 years.

The unifying insight across all segments: CSM's customers are not buying a product — they are buying continuity of operations, regulatory compliance, insurance coverage, and freedom from a catastrophic tail risk that existing solutions cannot address. The dielectric approach solves problems that Faraday shielding, surge protection, and isolation alone cannot solve.

SECTION 1: GOVERNMENT/DEFENSE PROCUREMENT OFFICERS

1.1 Persona: “Colonel Sarah Chen” — DOD Installation Resilience Director

Demographics - Age: 48-55 - Education: MS in Systems Engineering or Electrical Engineering; graduate of Defense Acquisition University (DAU) - Rank: O-5/O-6 (military) or GS-14/15 (civilian) - Agency: DOD Service Component (Army Corps of Engineers, NAVFAC, AFCEC), DTRA, or STRATCOM - Experience: 22-28 years in defense acquisition and infrastructure

Role and Responsibilities - Portfolio: \$200M-\$2B in military installation infrastructure projects annually - Decision authority: Can authorize programs up to \$50M; larger programs require SES/flag officer approval - Responsible for MILCON (military construction) EMP hardening requirements - Coordinates with combatant commands (STRATCOM, NORTHCOM) on EMP protection priorities - Reports to: SES-level Director of Installation Resilience (civilian) or Major General (military)

Pain Points 1. Enormous backlog of unhardened facilities — typical base has 200-800 structures, of which 5-15 are hardened. The gap is decades wide at current funding levels. 2. Traditional EMP hardening is extraordinarily expensive (\$3,000-\$10,000 per square foot for RF-shielded construction vs. \$200-\$400 for standard construction) 3. Hardening projects are scope-crept into irrelevance — what starts as an EMP project becomes a full facility renovation, delaying EMP protection by 3-5 years 4. Limited qualified contractors — only 15-25 US firms can bid on MILCON EMP hardening projects 5. Technology stagnation — “we’ve been hardening the same way since the 1980s”

Buying Process - Requirements development: 6-12 months (DD Form 1391 preparation) - Congressional authorization: Part of annual NDAA cycle - Congressional appropriation: Part of annual MILCON appropriation - Procurement method: Design-bid-build or design-build; typically full and open competition - Evaluation criteria: Technical capability (35%), past performance (25%), price (20%), small business participation (10%), schedule (10%) - Award timeline: 9-18 months from RFP release - Performance period: 18-36 months for construction

Decision Criteria (Ranked) 1. Certified EMP protection performance (MIL-STD-188-125 compliance) 2. Reduction in life-cycle cost vs. traditional hardening (maintenance eliminated) 3. Schedule acceleration (faster to build than traditional Faraday construction) 4. Operational flexibility (ability to modify facility in future without undermining EMP protection) 5. Past performance on DOD contracts 6. Price

Objections CSM Must Overcome 1. “We’ve never used dielectric materials for EMP protection. Where’s the

testing data?" 2. "MIL-STD-188-125 doesn't mention dielectric alternatives. How do we certify compliance?" 3. "Who installs this? Our existing contractors don't know dielectric construction." 4. "What's the 30-year sustainment plan? Will these materials last?"

CSM Acquisition Strategy - Provide MIL-STD-188-125 compliance testing data from Sandia National Laboratories - Develop UFC (Unified Facilities Criteria) language incorporating dielectric protection as an approved method - Partner with established DOD construction contractors (KBR, Fluor, AECOM, Bechtel) for installation - Offer 30-year material warranty backed by accelerated aging data from National Laboratory testing - Hire retired DOD installation engineers as CSM business development representatives

Lifetime Value: Single DOD installation hardening program: \$15-80M. Repeat business across multiple installations: \$100-500M over 10 years.

1.2 Persona: "Maria Rodriguez" — FEMA Regional Grants Administrator

Demographics - Age: 42-50 - Education: MPA or JD - Agency: FEMA Region (I-X) - GS Level: GS-14/15

Role: Administers BRIC and HMGP grant programs; reviews and approves state/local grant applications for infrastructure resilience projects. Manages \$200M-\$500M in annual grant awards per region.

Pain Points 1. Grant applicants don't understand EMP/CME risk — applications focus on floods, hurricanes, earthquakes 2. No standard cost-benefit analysis methodology for EMP/CME protection (BCA required for BRIC grants) 3. Political pressure to fund visible, popular projects (flood walls, fire stations) vs. invisible protection (EMP hardening) 4. Limited technical expertise to evaluate EMP protection proposals

CSM Acquisition Strategy - Develop and publish a FEMA-compliant BCA methodology for EMP/CME protection projects - Provide grant-writing support for state/local applicants proposing CSM equipment - Educate FEMA regional staff on EMP/CME risk and CSM's mitigation solution - Create template grant applications that reduce applicant burden for CSM-based projects

SECTION 2: ELECTRIC UTILITY EXECUTIVES

2.1 Persona: "James Thornton" — VP of Grid Resilience, Large IOU

Demographics - Age: 50-58 - Education: BS/MS in Electrical Engineering; MBA - Employer: Investor-Owned Utility (IOU) with 2-8 million customers - Reports to: SVP of Transmission & Distribution or Chief Operating Officer - Compensation: \$350K-\$600K (base + bonus); bonus tied to reliability metrics (SAIDI, SAIFI) and regulatory compliance

Role and Responsibilities - Portfolio: \$500M-\$1.5B in annual grid capital expenditure - Responsible for grid resilience strategy, including EMP/CME preparedness - NERC compliance officer for TPL-007 (GMD vulnerability assessments) - Manages relationship with state public utility commission (PUC) on resilience matters - Coordinates with regional reliability coordinators and ISOs/RTOs

Pain Points 1. TPL-007 GMD vulnerability assessments have identified 15-50 high-risk substations, but the corrective action budget covers only 1-3 per year 2. NERC compliance audits are increasingly focused on GMD preparedness — and regulators are unhappy with "study but don't act" posture 3. Long lead times for large power transformers (12-24 months) mean a GMD event that damages multiple transformers could cause 6-18 month outages 4. D&O liability concerns: board is asking "what are we doing about EMP?" 5. State PUC is unlikely to approve significant rate increases for EMP protection unless regulatory mandate exists 6. Engineering team is conservative — "we've been protecting substations with surge arresters and relays for 50 years; why do we need dielectric panels?"

Buying Process - Problem identification: TPL-007 assessment or NERC audit finding triggers action - Internal study: 6-12 months of engineering evaluation, technology assessment, vendor review - Regulatory pre-approval: 3-9 months of PUC engagement to ensure cost recovery - Procurement: 3-6 months RFP process; typically competitive bid - Award: Multi-year framework agreement preferred (3-5 years with renewal options) - Pilot: 1-2 substation pilot project before system-wide deployment (12-18 months) - Full rollout: 3-7 years to harden all identified critical substations

Decision Criteria (Ranked) 1. NERC compliance — does this satisfy TPL-007 corrective action plan requirements? 2. Proven reliability — what independent testing demonstrates protection effectiveness? 3. Cost recovery — will the state PUC allow rate-basing this investment? 4. Life-cycle cost — total cost of ownership over

30 years (including maintenance) 5. Operational compatibility — does this integrate with existing substation equipment and SCADA? 6. Vendor stability — will this company exist in 10 years to support the product?

Objections CSM Must Overcome 1. “Show me a utility that has already installed this.” 2. “We just spent \$2M on surge arresters. Why do we need dielectric panels too?” 3. “NERC hasn’t mandated dielectric materials. We’ll wait for the standard.” 4. “Our rate case is in 2028. We can’t add new capital projects until then.” 5. “What if your company goes bankrupt and we can’t get replacement panels?”

CSM Acquisition Strategy - Secure 2-3 pilot utility installations at zero cost (funded through DOE GRIP grants or CSM investment) to generate reference installations - Present at EEI (Edison Electric Institute), APPA (American Public Power Association), and NRECA (National Rural Electric Cooperative Association) conferences - Develop simplified NERC compliance documentation showing how CSM installation satisfies TPL-007 corrective action requirements - Partner with EPRI to conduct independent GMD protection performance testing - Offer “guaranteed rate recovery” — CSM will fund the installation if the PUC denies rate recovery, with repayment contingent on rate recovery approval - Provide 20-year system warranty backed by insurance (through CSM’s carrier partners)

Lifetime Value: Single large IOU: \$25-150M (initial hardening) + \$2-8M/year (maintenance, expansion, monitoring). 50 utilities: \$1.25-7.5 billion over 10 years.

2.2 Persona: “Robert Kipkemboi” — General Manager, Rural Electric Cooperative

Demographics - Age: 55-62 - Education: BS in Electrical Engineering or Business Administration - Employer: Rural electric cooperative serving 15,000-50,000 meters - Reports to: Cooperative Board of Directors (locally elected members) - Compensation: \$180K-\$280K

Differences from IOU Executive - Faster decision cycle — GM can authorize pilot projects with board approval (2-4 months vs. 18 months for IOUs) - More personal accountability — if the lights go out, the GM faces neighbors at the grocery store, not just regulators in a hearing room - Limited engineering staff — relies on NRECA and power supplier (G&T cooperative) for technical guidance - NRECA’s Cooperative Research Network (CRN) can fund pilot projects for member cooperatives

CSM Acquisition Strategy - Partner with NRECA to provide EMP/CME education for cooperative managers and boards - Develop “cooperative package” — simplified, pre-engineered dielectric protection for standard substation configurations at rural cooperatives - NRECA CRN funding for 5-10 cooperative pilot installations - Present at NRECA TechAdvantage and Regional Meetings

SECTION 3: INSURANCE UNDERWRITERS AND RISK MANAGERS

3.1 Persona: “Elizabeth Frost” — Chief Underwriting Officer, Large Commercial Property Insurer

Demographics - Age: 45-52 - Education: BS in Actuarial Science, Risk Management, or Mathematics; CPCU designation; often MBA - Employer: Major commercial property insurer (AIG, Chubb, Zurich, AXA XL) - Reports to: Chief Underwriting Officer or Chief Risk Officer - Compensation: \$400K-\$800K; bonus tied to combined ratio and portfolio profitability

Role and Responsibilities - Portfolio: \$2-8 billion in commercial property premiums - Authority: Can authorize new product lines, policy wording changes, and underwriting guidelines - Responsible for aggregate portfolio risk management, including emerging perils - Reports to board risk committee on emerging risk exposure

Pain Points 1. Aggregate concentration risk from “silent” EMP/CME exposure — policies don’t exclude it, so the entire portfolio is exposed 2. Catastrophe models don’t include space weather — can’t quantify exposure, can’t price it, can’t manage it 3. Reinsurers are beginning to ask about EMP/CME exposure in treaty renewals 4. ORSA (Own Risk and Solvency Assessment) filings increasingly require identification of non-modeled catastrophic risks 5. Board risk committee is asking “what keeps you up at night?” and EMP/CME is emerging as the answer

Buying Process - Internal risk assessment: 3-6 months of exposure quantification (or acknowledgment that exposure can’t be quantified) - Reinsurance consultation: 3-6 months of discussion with treaty reinsurers about EMP/CME sublimits or exclusions - Policy wording review: 6-12 months of legal review to add EMP/CME exclusions or hardening requirements - Regulatory filing: 3-9 months for state insurance department approval of new policy forms - Implementation: 12-24 month rollout to existing policyholders at renewal

Decision Criteria (Ranked) 1. Does this reduce aggregate portfolio EMP/CME exposure by a measurable amount? 2. Is the hardening system independently tested and certified? 3. What is the premium reduction justified by the reduced exposure? (actuarial soundness) 4. Will reinsurers accept this as a valid risk mitigation for treaty purposes? 5. What is the administrative burden of implementing a hardening-credit program?

CSM Acquisition Strategy - Provide CSM-developed EMP/CME exposure model for commercial property portfolios (insurers need this first) - Demonstrate premium-reduction economics using insurer-specific portfolio data - Offer reinsurance-backed warranty on CSM system performance (removes insurer's technology performance risk) - Develop turnkey "Resilience Credit Program" — CSM handles the hardening specification, installation, inspection, and monitoring; the insurer simply offers the premium credit - Present at RIMS, PCI (Property Casualty Insurers), and NAIC conferences

Lifetime Value: Single insurer partnership: \$50-200M in CSM system sales to policyholders over 5 years. 10-15 insurer partnerships: \$500M-\$3B over 5 years.

3.2 Persona: "Thomas Bjornsen" — Reinsurance Treaty Underwriter, Swiss Re / Munich Re

Demographics - Age: 40-48 - Education: PhD in Physics, Mathematics, or Catastrophe Modeling - Employer: Major reinsurer (Swiss Re, Munich Re, Hannover Re, SCOR) - Compensation: \$300K-\$500K; bonus tied to treaty portfolio profitability

Role: Underwrites reinsurance treaties for primary insurers; assesses aggregate risk across cedants' portfolios; sets terms, conditions, and pricing for reinsurance coverage.

Pain Points 1. Primary insurers' portfolios contain unquantified EMP/CME exposure that is being ceded (unknowingly) to reinsurers 2. No treaty language specifically addresses EMP/CME risk — it's "silently" included in all-property treaties 3. A Carrington-class event would trigger simultaneous claims across all 50+ cedants in a reinsurer's portfolio — the ultimate accumulation scenario 4. Rating agencies (S&P, AM Best, Moody's) have not yet incorporated space weather into reinsurer rating criteria, but this is expected to change

CSM Acquisition Strategy - Provide accumulation risk modeling showing how CSM hardening reduces aggregate reinsurer exposure - Offer parametric EMP/CME product that is reinsurer-backed (provides new premium pool for reinsurers) - Educate treaty underwriters on space weather risk and the differential impact on hardened vs. unhardened portfolios - Publish research in reinsurance industry journals (Geneva Association, Casualty Actuarial Society)

SECTION 4: MARITIME FLEET OPERATORS

4.1 Persona: "Captain Henrik Larsen" — VP of Technical Operations, Major Container Line

Demographics - Age: 50-58 - Education: Master Mariner certification; MBA or MS in Marine Engineering - Employer: Major container shipping line (Maersk, MSC, CMA CGM, Hapag-Lloyd) or tanker operator - Reports to: Chief Operating Officer or Chief Technical Officer - Compensation: \$350K-\$600K

Role and Responsibilities - Fleet: 50-500 vessels - Responsible for vessel technical management, maintenance, retrofitting, and regulatory compliance - Manages \$200M-\$1B annual technical budget - Coordinates with classification societies (Lloyd's Register, DNV, ABS, Bureau Veritas) on vessel certification

Pain Points 1. Modern vessels are floating data centers — integrated bridge systems, ECDIS, DP systems, satellite comms — all EMP-vulnerable 2. A vessel disabled at sea by EMP loses navigation, communication, and propulsion control; salvage costs exceed \$10M per incident (BIMCO data) 3. Classification societies are developing cyber/electronic warfare resilience standards that will apply to newbuilds by 2028-2032 4. Hull and machinery insurance is silent on EMP/CME; unclear whether disabled-vessel claims would be paid 5. Cargo owners are beginning to ask about supply chain resilience, including EMP/CME

Buying Process - Fleet assessment: 6-12 months of vulnerability analysis for critical navigation and control systems - Classification society consultation: 3-6 months of engagement with class on protection standards - Newbuild specification: EMP protection integrated into newbuild contracts at design phase (18-24 months before delivery) - Retrofit program: 3-7 year rolling program for existing fleet retrofit - Insurance coordination: coordination with marine underwriters on premium implications

Decision Criteria (Ranked) 1. Classification society acceptance — will class approve this as meeting resilience requirements? 2. Weight — every pound of protection reduces cargo capacity. Dielectric advantage is compelling here (60-85% weight reduction vs. Faraday shielding) 3. Maintenance — no corrosion; reduced dry-dock time 4. Retrofit-ability — can this be installed during a standard 2-week dry-dock period? 5. Insurance premium impact — will marine underwriters reduce premiums for hardened vessels?

CSM Acquisition Strategy - Engage IACS member classification societies to develop dielectric protection certification standards - Pilot installation on 2-3 vessels with a major container line (cost shared with insurer) - Develop modular, pre-fabricated dielectric panel system for standard vessel bridge configurations - Present at maritime industry conferences (CMA Shipping, Nor-Shipping, Posidonia) - Partner with major shipyards (Hyundai Heavy, Daewoo, Samsung Heavy, Fincantieri, Meyer Werft) for newbuild integration

Lifetime Value: Single major shipping line: \$50-200M (fleet retrofit). Top 10 shipping lines: \$500M-\$2B over 10 years.

SECTION 5: HOME BUILDERS AND DEVELOPERS

5.1 Persona: “Michael Torres” — President, Regional Luxury Home Builder

Demographics - Age: 45-55 - Education: BS in Construction Management; MBA - Company: Regional luxury home builder constructing 50-200 homes/year at \$800K-\$3M price points - Geography: Texas, Florida, California, Colorado — states with EMP/CME legislation or high awareness

Role and Responsibilities - Manages construction operations, supplier relationships, and customer sales - Differentiates homes through features, quality, and technology integration - Responsible for builder profitability (15-25% gross margin target)

Pain Points 1. Luxury home buyers increasingly ask about “resilience” — backup power, hardened construction, self-sufficiency (Coldwell Banker survey: 37% cite resilience as top-3 priority) 2. High-net-worth insurance market is hardening — premium increases of 20-40% in catastrophe-exposed areas; anything that reduces premiums is valuable 3. Smart home systems (Control4, Crestron, Savant) represent \$50K-\$250K in electronics that are EMP-vulnerable 4. Competitive differentiation is difficult in luxury home market — everyone has granite countertops and Wolf ranges

Buying Process - Product evaluation: Builder evaluates CSM’s Aegis Home system (1-3 months) - Model home installation: One model home with Aegis Home system (2-3 months) - Customer offering: Aegis Home offered as optional upgrade or standard feature (dependent on price point) - Customer decision: 30-90 day sales cycle with individual home buyers

Decision Criteria (Ranked) 1. Customer willingness to pay — can this be sold as an upgrade, or must it be standard? 2. Insurance premium reduction — can homeowners recoup the cost through lower premiums? 3. Installation complexity — can the builder’s existing electrical contractor install this? 4. Warranty and support — what’s the warranty period, and who handles service calls? 5. Marketing support — will CSM provide co-marketing materials for builder sales centers?

CSM Acquisition Strategy - Pilot with 2-3 regional luxury builders in high-awareness markets (Texas, Florida) - Co-market with Chubb, AIG Private Client Group, PURE Insurance — “Aegis Home certified” homes receive premium reduction - Develop simplified installation process for electrical contractors (pre-fabricated panels with plug-in connections) - Create builder certification program (“Aegis Home Certified Builder”) with training, marketing support, and lead generation - Offer 10-year system warranty, serviceable by CSM-certified electricians

Lifetime Value: Single builder relationship (200 homes/year × 10 years): \$25M. 50 builder relationships: \$625M over 10 years.

SECTION 6: INFRASTRUCTURE ENGINEERING FIRMS

6.1 Persona: “Dr. Anika Patel” — Principal Engineer, Global Infrastructure Consulting Firm

Demographics - Age: 48-55 - Education: PhD in Civil/Structural/Electrical Engineering; PE license in multiple

states - Firm: AECOM, WSP, Jacobs, Stantec, Arup, or Burns & McDonnell - Reports to: Practice Lead for Resilience or VP of Energy - Compensation: \$250K-\$400K

Role: Specifies protection systems in infrastructure designs; influences procurement decisions for substations, data centers, hospitals, and government facilities designed by the firm. The specifier — not the buyer — often determines what protection technology is used.

Pain Points 1. Clients ask for “EMP protection” but there’s no standard specification to reference — every project is custom-engineered 2. Traditional Faraday shielding creates conflicts with other design requirements (ventilation, access, aesthetics, structural load) 3. Limited qualified contractors for EMP shielding installation — creates schedule risk 4. Liability exposure if a design they specify fails under EMP/CME conditions

CSM Acquisition Strategy - Develop CSI (Construction Specifications Institute) format specifications for dielectric EMP protection (enables specifiers to easily reference CSM in project documents) - Provide continuing education (CEU/PDH) courses for engineers on dielectric EMP protection - Offer design support services — CSM engineers collaborate with firm’s designers on specific projects - Co-present at ACEC (American Council of Engineering Companies), ASCE, and IEEE conferences - Develop BIM (Building Information Modeling) objects for CSM dielectric panels for integration into design software (Revit, AutoCAD)

SECTION 7: CROSS-SEGMENT BUYING TRIGGERS

Understanding what causes each segment to initiate a purchase is critical for targeting CSM’s sales and marketing efforts:

Trigger	Government	Utilities	Insurance	Maritime	Home Builders	Engineers
Regulatory mandate (FERC NOPR, CER Directive)	★★★★★	★★★★★	★★★★☆	★★★★☆	★★☆☆☆	★★★★★
NERC compliance audit finding	★★★★☆	★★★★★	★★☆☆☆	☆☆☆☆☆	☆☆☆☆☆	★★★★☆
Insurance non-renewal / premium spike	★★☆☆☆	★★★★☆	★★★★★	★★★★☆	★★★★★	★★☆☆☆
D&O liability concern	★★★★☆	★★★★☆	★★★★☆	★★☆☆☆	★★☆☆☆	★★★★☆
Competitive pressure (peer adoption)	★★★★☆	★★★★☆	★★★★☆	★★★★☆	★★★★★	★★★★☆
Grant availability (FEMA BRIC, DOE GRIP)	★★★★★	★★★★☆	★★☆☆☆	★★☆☆☆	★★☆☆☆	★★★★☆
Catastrophic near-miss event	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★
Classification society mandate	☆☆☆☆☆	☆☆☆☆☆	★★☆☆☆	★★★★★	☆☆☆☆☆	★★★★☆
Technology refresh cycle	★★★☆☆	★★★☆☆	★★☆☆☆	★★★☆☆	★★☆☆☆	★★★☆☆

SECTION 8: CUSTOMER ACQUISITION COST BY SEGMENT

Segment	CAC Estimate	Sales Cycle	Deal Size (Typical)	CAC Payback	Acquisition Channel
					Direct sales;

Government (DOD)	\$250K-\$750K	18-36 months	\$5M-\$80M	<6 months	SBIR; government relations
Government (FEMA)	\$150K-\$400K	12-24 months	\$2M-\$15M	<6 months	Grant support; direct sales
Utility (Large IOU)	\$200K-\$500K	18-30 months	\$10M-\$100M	<3 months	Direct sales; industry conferences; EPRI NRECA partnership; regional workshops
Utility (Cooperative)	\$50K-\$150K	6-12 months	\$500K-\$3M	<6 months	Executive relationships; RIMS conference
Insurance (Carrier)	\$500K-\$2M	24-36 months	\$50M-\$200M (via policyholders)	<3 months	Classification society introductions; direct sales
Maritime (Major Line)	\$200K-\$500K	18-30 months	\$20M-\$100M	<3 months	Builder partnerships; home shows
Home Builder	\$50K-\$150K	6-12 months	\$2M-\$25M (via home buyers)	<6 months	ACEC/IEEE conferences; CEU courses
Engineering Firms	\$75K-\$200K	6-18 months	Creates channel, not direct revenue	N/A	

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